COMPOSITION IN A GROUPOID OVER A FIELD IS FLAT AND UNIVERSALLY OPEN

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- The pull diagram isomorphism. equation-pull (same chapter, structural anchor). Identifies c with a projection pr_1 : R x_{t,U,t} R -> R.
- Universal openness over fields. Morphisms of Spaces, lemma space-over-field-universally-open (cross-chapter, named). Any morphism from a space over a field to the field base is universally open.

Source reference. Original equation 06DZ: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 870–870. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 03C7: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 303–303. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06DX: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 853–853. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06DY: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 859–859. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06DN: focused-seeds/geometry/S9-smg-groupoid-on-field-open-multiplication/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-more-groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *In Situation 06DY the composition morphism $c : R \times_{s,U,t} R \to R$ is flat and universally open. In Situation 06DX the group law $m : G \times_k G \to G$ is flat and universally open.*

Proof. The composition is isomorphic to the projection map $\mathrm{pr}_1 : R \times_{t,U,t} R \to R$ by Diagram (03C7). The projection is flat as a base change of the flat morphism t and open by Morphisms of Spaces, Lemma 06DN. The second assertion follows immediately from the first because m matches c in (06DZ). □